AP CALCULUS AB

Unit 1 • Self Check 1.1 and 1.2 • Lessons 1-1 to 1-4

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Name: _____ Date: _____

Directions

Answer every question. Multiple-choice questions have one best answer. On free-response questions, show your work and justify your answers. Questions labeled **NO CALCULATOR** must be done without a calculator; questions labeled **CALCULATOR** require a graphing calculator. For questions with a graph, read the values you need from the grid.

Question 1

NO CALCULATOR

$$\lim_{x \to -3} \frac{x^2 + x - 6}{x^2 - 9} =$$

(A) $\frac{5}{6}$

(B) 0

(C) $-\frac{5}{6}$

(D) It does not exist.

Question 2

NO CALCULATOR

$$\lim_{x \to \infty} \frac{x^2}{e^x} =$$

(A) ∞

(B) 1

(C) e

(D) 0

Question 3

NO CALCULATOR

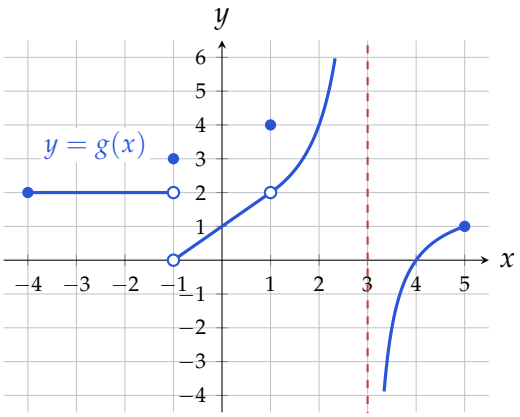
$$\lim_{x \to -1} \frac{x^2 - 1}{|x + 1|}$$

(A) 2

(B) -2

(C) 0

(D) does not exist



Question 4 NO CALCULATOR

Using the graph of g, at which x is the discontinuity removable?

(A) $x = -1$
(B) $x = 3$
(C) $x = -4$
(D) $x = 1$

Question 5 (Free response) NO CALCULATOR

Using the graph of g, classify the discontinuities at $x = -1$, 1 and 3, with limit statements.